



UNIVERM TOTAL Tablets A.U.V.

Name and Strength of Active Substance(s)

50 mg Praziquantel, 144 mg Pyrantel Embonate, 200 mg Fenbendazole

Product Description

Yellow or yellowish-grey tablets, of disc form with cut edges, bisectrix on one side. Odd stains, discolourations or cracks must not be found on the tablets. The surface should be smooth.

Pharmacodynamics

Mechanism of action.

Pyrantel, a tetrahydropyrimidine derivative and its embonate salt, has been a well-known wide-spectrum anthelmintic against the parasites in dogs and other animals since 1966. Pyrantel depolarises neuromuscular synapses. It also blocks the choline esterase enzyme. These cell biological changes cause a paralysis of the worms, and thus their death. Pyrantel embonate component is effective against gastrointestinal roundworms and tapeworms.

Praziquantel, an isoquinolin derivative, is a highly effective anthelmintic against a wide range of tapeworms in adult and late immature forms in humans and animals alike. Praziquantel is very rapidly absorbed through the parasite's surface and distributed throughout the parasite. Praziquantel causes severe damage to the parasite integument, resulting in the contraction and paralysis of the parasites. There is an almost instantaneous tetanic contraction of the parasite musculature and a rapid vacuolization of the syncytial tegument. This rapid contraction has been explained by changes in divalent cation fluxes, especially calcium.

Fenbendazole, a metyl-5(fenilitio)-2-benzimidazol carbamate, is a widely used benzimidazole compound (pharmacologically active metabolite of febantel). Mechanism of action is based on inhibition of polymerization of microtubules. Fenbendazole blocks the fumarate reductase enzyme in the worms, and blocks the glucose uptake, thus leading to damage of their energy-producing metabolic processes. Upon exhaustion of its energy reserves cause the paralysis of the worms, and thus their death. It is suitable for a treatment against tapeworms, hookworms, whipworms, lungworms *Cysticercus tenuicollis* and *Giardia* infections.

Synergy of the combination of pyrantel embonate and fenbendazole ensures an increased effectivity against whipworms. Pyrantel embonate component alone ensures higher effectivity against the nematodes than fenbendazole, except whipworms, which proves the necessity of pyrantel embonate in the product.

Although fenbendazole is effective against *Taenia pisiformis* to some extent, the presence of praziquantel is still necessary in the combination in order to achieve 100% effectivity in the treatment against *T. pisiformis*, and *D. caninum*. There is evidence that fenbendazole has only a slight effect against *D. caninum* and *T. pisiformis*. The product is a wide-range anthelmintic. It is effective against the following species: *Toxocara canis*, *Toxascaris leonina*, *Ancylostoma caninum*, *Uncinaria stenocephala*, *Trichuris vulpis*, *Dipylidium caninum*, *Taenia multiceps* and *Mesocostoides species*. All ingredients of the product have an antiparasitic effect based on a different mechanism. This is particularly important because parasites might become resistant to certain anthelmintics after a while.

Pharmacokinetics

Perorally administered praziquantel is absorbed almost completely from the intestinal tract. Maximum plasma concentration is obtained after 30-60 minutes in dogs. After absorption the drug is distributed to all organs; it passes the blood-brain barrier and also passes into the bile. Praziquantel is metabolized into inactive forms in the liver and secreted in bile. Inactive metabolites of praziquantel are excreted in the urine. The ubiquitous distribution is an asset in activity of praziquantel against cestodes larval or adult forms that have highly varied locations in the host. Praziquantel has rapid metabolism and excretion in dogs.

The embonate salt of pyrantel has low aqueous solubility, an attribute that reduces absorption from the gut and allows the drug to reach and be effective against parasites in the large intestine. Following absorption, pyrantel embonate is quickly and almost completely metabolized into inactive metabolites that are excreted rapidly in the urine and faeces. Maximum plasma concentration is obtained after 4-6 hours via oral administration in dogs.

Fenbendazole is a benzimidazole-compound and active metabolite of febantel.

Fenbendazole and its known active sulphoxide metabolite in dogs appear in the liver and excreted via urine and faeces. Maximum plasma concentration is realised between the first 1-2 hours after oral administration of the drug to dogs. 24-48 hours after this interval, the active ingredient cannot be identified at the plasma level.

Environmental Properties

Not relevant

Indication

For the treatment of mixed infections by both adult nematodes and cestodes in dogs of the following species:

Ascarids: *Toxocara canis*, *Toxascaris leonina* (adult and late immature forms)

Hookworms: *Ancylostoma caninum*, *Uncinaria stenocephala* (adults)

Whipworms: *Trichuris vulpis* (adults)

Tapeworms: *Dipylidium caninum*, *Taenia hydatigena*, *Taenia pisiformis*. (adult and late immature forms)

Recommended Dosage

Dosage: For oral treatment only.

The recommended dose is 5 mg/kg praziquantel, 14,4 mg/kg pyrantel embonate and 20 mg/kg fenbendazole (this is equivalent to 1 tablet/ 10 kg bw). For routine treatment a single dose is sufficient.

In case of a diagnosed helminthiasis, the treatment should be repeated in 14 days.

To ensure administration of a correct dose, body weight should be determined as accurately as possible. Dosage should be established by the veterinarian.

Dog's bodyweight (kg)	Amount of tablets (piece)
Puppies and small dogs	
2-5	1/2
5-10	1
Medium size dogs	
10-20	2
20-30	3
Large dogs	
31-40	4





Method of administration:

The tablet can be given directly for oral administration or crushed, mixed into the feed.

The animal not needs to be starved during the treatment.

In cases of confirmed mono-infections by cestodes or nematodes, a monovalent product containing a cestocide or a nematocide alone should be used.

Mode of Administration

Orally

Contraindications

Do not use in case of hypersensitivity to the active substances or to any of the excipients.

Do not use simultaneously with any product containing piperazin derivatives and/or organic phosphate ester.

Warnings and Precautions

Pyrantel should be given with caution for dogs with cachexia. Parasite resistance to any particular class of anthelmintic may develop following frequent, repeated use of an anthelmintic of that class. Fleas serve as intermediate hosts and source of infection for one common type of tapeworm – *Dipylidium caninum*. Tapeworm infestation may reoccur unless control of intermediate hosts as well as the environment is undertaken concurrently to the treatment.

Special precautions for use in animals: It is recommended to keep the dogs confined for 24 hours after treatment and dispose the excreted faeces, worms, segments and eggs. It is recommended to clean and disinfect often the environment of the animals. In debilitated or heavily infested animals, the product should be used only after evaluation of the risk/benefit by the veterinarian. Treatment of animals less than 6 weeks old may not be necessary. Follow-up examinations of faeces or retreatment with a nematocidal product may be needed in case of *Ancylostoma caninum* or *Toxocara canis* infections.

Special precautions to be taken by the person administering the veterinary medicinal product to animals: In case of accidental ingestion, seek medical advice immediately and show the package leaflet or the label to the physician.

People with known hypersensitivity to praziquantel, pyrantel pamoate or fenbendazole should avoid contact with the veterinary medicinal product. Wash hands after use.

Keep children away from dogs the days after treatment.

Interactions with Other Medicaments

Do not use simultaneously with piperazine compounds as the anthelmintic effects of pyrantel and piperazine may be antagonized. Equally, do not administer together with organic phosphate esters and diethylcarbamazine. Due to similar mechanism of action and toxicological characteristics do not use this anthelmintic simultaneously with morantel and combinations of morantel.

Statement on usage during pregnancy and lactation

The safety of the veterinary medicinal product has not been established during the early pregnancy in dogs. Do not use in pregnant bitches during the first four weeks of gestation. After this time and during lactation use only according to the benefit risk assessment by the responsible veterinarian. In breeding dogs use only according to the benefit risk assessment by the responsible veterinarian.

Adverse Effects/Undesirable Effects

Rarely, a lack of appetite, diarrhoea, vomiting, and distress or, transitionally, a rise in the level of AST (aspartate aminotransferase) might occur.

Overdose and Treatment

In target animal tolerance studies, no clinical, haematological or biochemical parameters have changed significantly, even at doses in excess 5 times the recommended dose for 3 days. Overdose of this medicinal product can occur very rarely under field conditions because of its wide therapeutic range. If suspected toxic reactions do occur due to extreme overdose, the appropriate symptomatic treatment should be initiated if necessary.

Withdrawal Period(s)

Not applicable.

Storage Conditions

To be stored less than 25 °C, in a dry place. Keep away from light.

Proposed Shelf Life

3 years.

Dosage Forms

Tablet.

Packaging available

20 tablets a box.

Name and Address of Manufacturer

Pernix Pharma Gyogyszergyarto Kft., Hungary.

Name and Address of Product Registration Holder

Taseen Trading Sdn. Bhd. ,

No 8, Jalan Astana 1F/KU2, Bandar Bukit Raja, 41150 Klang, Selangor, Malaysia.

Date of Revision of Package Insert

14/02/2019